

Troubleshooting Cisco Catalyst 9000 Series Switches

cisco Live !

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Agenda

- 01 Introduction
- 02 Platform
- 03 Packet Visibility
- 04 Packet Drops
- 05 Forwarding Verification



Introduction

Catalyst 9k family



Format	Stackable	Stackable	Chassis	Standalone	Chassis
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9350, 9500X and 9600 Sup II based on Silicon One
All others based on UADP Asics



Different models designed for
different uses and performance



Catalyst IOS-XE Software release schedule

	Cupertino 17.12	Dublin 17.15	17.17	17.18	26.1.1
Next planned release	17.12.7	17.15.4	None	17.18.3	26.1.1
End of Software Maintenance	March 2026	March 2027	July 2026	-	-
End of Vulnerability Security Support	September 2027	September 2028	July 2026	-	-
Extended Maintenance Release	Yes	Yes	No	Yes	No

Recommended releases:

<https://www.cisco.com/c/en/us/support/docs/switches/catalyst-9300-series-switches/214814-recommended-releases-for-catalyst-9200-9.html>

Software Maintenance Upgrade(SMU)

- SMU's provide pointed fix on top of IOS-XE and are released for critical defects
 - Hitless , No reload needed to apply on top of IOS-XE
 - Reboot, Reload is performed to apply SMU
- Installation :
 - install add file <smu> activate commit
 - or
 - install add file <smu>
 - install activate file <smu>
 - install commit
- Confirmation:
 - show install summary

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Select a Software Type

[IOS XE Hardware Programmable Devices](#)
[IOS XE Software](#)
[IOS XE Software AP Device Pack](#)
[IOS XE Software Maintenance Upgrade AP Service Pack \(SMU APSP\)](#)
[IOS XE Software Maintenance Upgrades \(SMU\)](#)
[IOS XE Wireless Controller Software Package](#)
[NBAR2 Protocol Packs](#)
[NBAR2 Protocol Packs Signed](#)
[Security Sensor](#)

Install Summary

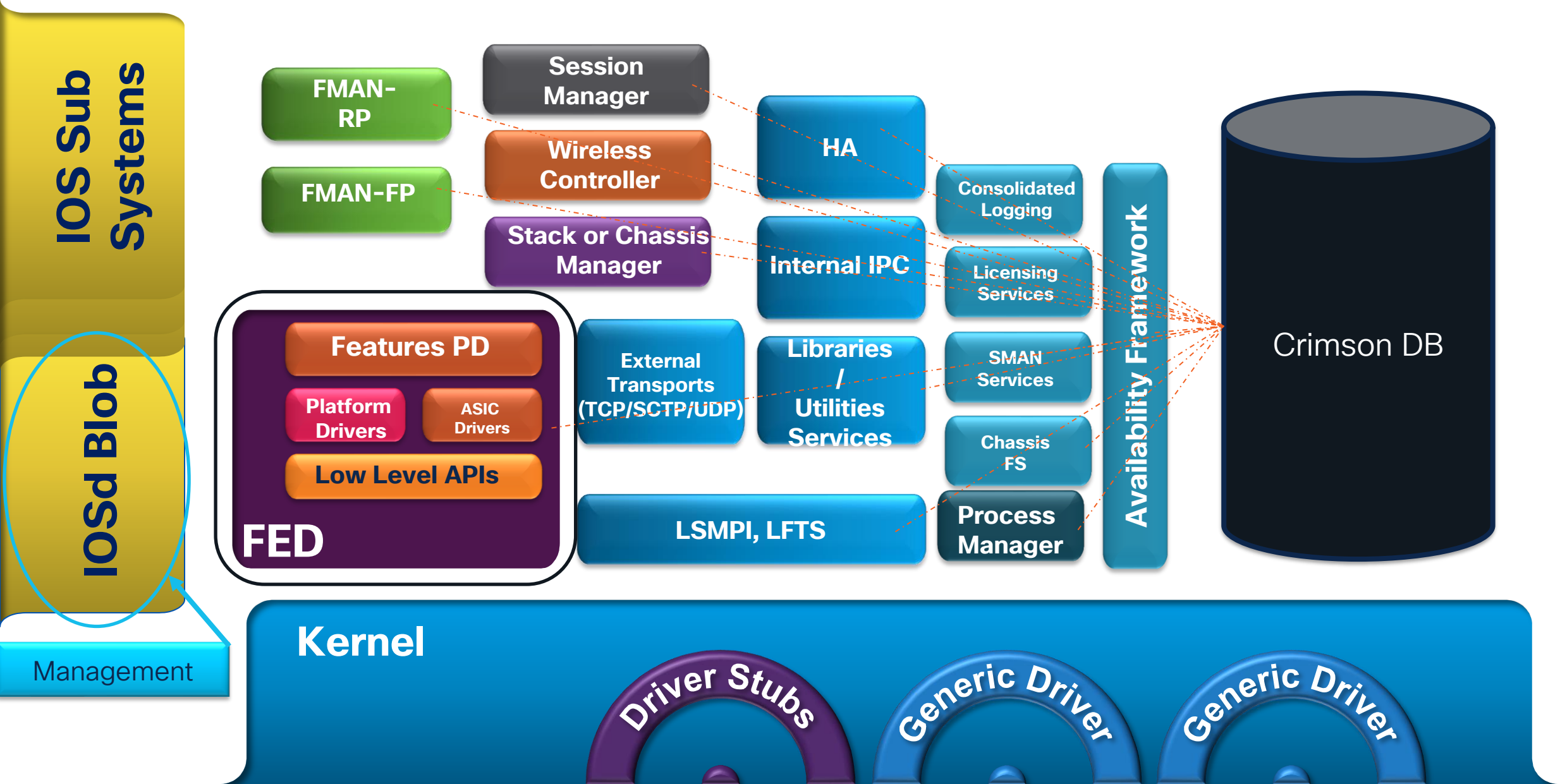
```
9400_21#show install summary
[ R0 R1 ] Installed Package(s) Information:
State (St): I - Inactive, U - Activated & Uncommitted,
             C - Activated & Committed, D - Deactivated & Uncommitted
-----
Type  St   Filename/Version
-----
IMG   C    17.12.04.0.4826
PKG   C    bootflash:C9800-SW-iosxe-wlc.17.12.04.SPA.bin
SMU   C    bootflash:cat9k_iosxe.17.12.04.CSCwk22887.SPA.smu.bin
SMU   D    bootflash:cat9k_iosxe.2025-01-29_05.38.0.CSCwn37241.SSA.smu.bin
```

- SMU's should be in state Activated and Committed (C)
- Show install summary also shows packages (WLC) and base image (IMG)
- .SPA. : Production images
.SSA. : Special or Engineering Images.

Internetworking Operating System – eXtended Edition.

- IOS-XE runs on top of a Linux Kernel allowing multi-core architectures
- IOSd runs as a process on top of the kernel ,still retains its own scheduling
- Some IOS-XE features have been taken out of IOSd and given their own process, forwarding manager, session manager, wireless processes
- Catalyst 9000 run 16.x , 17.x, 21.x versions
- Catalyst 9000 Platform Specific code is running in the Forwarding Engine Driver Processes (FED)
- Both S1 and UADP processes utilize FED
- Management interface is through IOSd processes

IOS-XE graphical overview



Various level of CLI commands

Platform Independent CLI

```
show ip route  
show version
```

IOS

IOS-XE Specific CLI

```
show platform software ip switch active rp cef  
show platform software ip rp active cef
```

IOS-XE

Catalyst 9000 Platform Specific CLI

```
show platform software fed switch 5 ifm mappings  
show platform software fed active ifm mappings
```

Platform

CLI Commands , location

Show platform commands can be executed on different locations.

Different Supervisor Module:

- Active
- Standby

Different Stack Member:

- switch active
- switch standby

Same module/switch:

- Forwarding Processors (F0/F1)
- Routing Processors (R0/R1)

Depending on the location the command is executed the output will vary.

Debugging challenges

- Traditional Debugs can be CPU intensive
- IOSd process has no direct visibility to the operation of other processes running on IOS-XE
- The solution on IOS-XE Binary Tracing
- Binary Tracing is always on tracing per IOS-XE process
- Common processes to enable tracing :
 - smd , Session Manager (Authentication)
 - wncd, Wireless Process
 - IOSRP , IOSd Process
 - fman , Forwarding Manager process (_fp and _rp)
 - fed, Forwarding Engine Driver , Catalyst 9k platform specific process

Setting Trace Levels

- By Default, trace levels of all processes for all facilities is set to Notice
- Per process the facilities can be set to a higher or lower level

debug	Debug messages
emergency	Emergency possible message
error	Error messages
info	Informational messages
noise	Maximum possible message
notice	Notice messages
verbose	Verbose debug messages
warning	Warning messages

Setting all other facilities to emergency level helps with noise in trace

```
Switch#set platform software trace smd switch active R0 all emergency
Switch#set platform software trace smd switch active R0 dot1x-all verbose
Switch#show platform software trace level smd switch active R0 | inc dot1x
dot1x                                     Notice
dot1x-all                               Verbose
dot1x-redun                             Notice
Switch#set platform software trace all notice ← set all back to default
```


Displaying trace logs

- Tracelog files are stored in crashinfo:/logs in binary format.
- Analyzing traces: show logging process <process>
- Bundle creation of binary traces: “request platform software trace archive”
- Appending command with internal *could* provide more detail

```
Edge_1#sh logging process smd | inc RADIUS
2022/06/06 23:24:03.268912 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: Send Accounting-Request to
10.48.91.222:1813 id 1813/184, len 850
2022/06/06 23:24:03.268937 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: authenticator e5 d1 b7 4d 8b e9 d5
06 - 14 b9 8d b6 8c 29 93 94
2022/06/06 23:24:03.268945 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: Vendor, Cisco [26] 211
2022/06/06 23:24:03.268954 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: Cisco
AVpair [1] 205 "cts-pac-opaque="
2022/06/06 23:24:03.268960 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: Vendor, Cisco [26] 36
2022/06/06 23:24:03.268966 {smd_R0-0}{1}: [radius] [24732]: (info): RADIUS: Cisco
AVpair [1] 30 "dc-profile-name=Cisco-Device"
```

Make sure to enable traces in the right places.

Platform

Looking at the Kernel CPU information

- Each Supervisor and stack member runs its own Kernel

```
9300_54#show proc cpu platform sorted location switch active R0
CPU utilization for five seconds: 12%, one minute: 9%, five minutes: 5%
Core 0: CPU utilization for five seconds: 14%, one minute: 10%, five minutes: 5%
Core 7: CPU utilization for five seconds: 14%, one minute: 10%, five minutes: 6%
  Pid      PPid      5Sec      1Min      5Min  Status      Size  Name
-----
 14854    13498    51%     38%     22%   S          951436  linux_iosd-imag
 15322    15160    42%     30%     15%   S          313236  fed main event
   2960         2     8%      5%      2%   R           0  lsmpi-xmit
   2961         2     5%      4%      2%   S           0  lsmpi-rx
```

- linux_iosd-image process as seen on kernel is the IOSd process
- Lsmpi processes are part of Linux Shared Memory Punt Interface
- Processes running might vary depending on if it's Active/Standby or Member

IOSd CPU utilization

```
9300_54#show proc cpu sort
```

```
CPU utilization for five seconds: 25%/1%; one minute: 22%; five minutes: 11%
```

PID	Runtime(ms)	Invoked	uSecs	5Sec	1Min	5Min	TTY	Process
40	9474304	54714487	173	22.95%	16.12%	8.15%	0	ARP Input
24	95502	12431773	7	0.07%	0.00%	0.00%	0	IPC Mcast Pendin
41	152951	13280277	11	0.07%	0.00%	0.00%	0	ARP Background
53	102409	12729984	8	0.07%	0.00%	0.00%	0	Dynamic ARP Insp
83	4759850	48620416	97	0.07%	0.03%	0.02%	0	IOSD ipc task
121	5602021	50896336	110	0.07%	0.04%	0.05%	0	Crimson config p
136	5112131	792052381	6	0.07%	0.03%	0.02%	0	L2 LISP Punt Pro

- Inside IOSd process many functions and processes run
Examples: BGP, RIP, CEF, ARP, UDLD, CDP, SSH, SNMP, Telnet
- Data plane traffic is forwarded by S1/UADP asic's without CPU involvement
- Control Plane Policing enabled by default to throttle traffic

Catching CPU spikes in IOSd

- CPU spikes often occur when not actively watching the switch.
- IOS allows CPU monitoring :
process cpu threshold type total rising <%> interval <s> falling <%> interval <s>
- When threshold are exceeded syslog is generated
- Syslog can be used by Embedded Event Manager to automate collection of data during time of failure

```
%SYS-1-CPURISINGTHRESHOLD: Threshold: Total CPU Utilization(Total/Intr): 48%/15%, Top 3 processes (Pid/Util): 40/32%, 94/0%, 105/0%
```

```
9300_54#show proc cpu | inc 40 | 94 | 105 | PID
```

Top 3 processes,
ARP input is highest

PID	Runtime(ms)	Invoked	uSecs	5Sec	1Min	5Min	TTY	Process
40	19640520	69264320	283	22.48%	20.61%	10.14%	0	ARP Input
94	652410	6428606	101	0.07%	0.00%	0.00%	0	PuntInject Keepa
105	3477489	6127722	567	0.00%	0.02%	0.02%	0	Crimson flush tr

Unexpected Reloads

- Unexpected reloads or software exception occur for a variety of reasons.
- Switchovers don't always get noticed, confirm switch uptimes

```
R10_BN_19#show version
..
R10_BN_19 uptime is 4 weeks, 4 hours, 2 minutes
Uptime for this control processor is 2 hours, 9 minutes
System returned to ROM by Critical software exception,
system report at /crashinfo/R10\_BN\_19\_1\_RP\_0-system-report\_1\_20260211-123051-UTC.tar.gz
```

- In redundant system also check uptime of all switches and supervisor's
- The switch or supervisor that currently active might not always have been

```
Switch#show version | inc uptime
Switch uptime is 7 weeks, 2 days, 4 hours, 43 minutes
Switch uptime           : 2 weeks, 3 days, 2 hours, 13 minutes
Switch uptime           : 7 weeks, 2 days, 4 hours, 45 minutes
```

Reload History

- Use the logging onboard command to get a history of reloads

```
R10_BN_19#sh logging onboard switch active uptime detail
```

```
-----
UPTIME SUMMARY INFORMATION
-----
First customer power on : 01/30/2019 05:20:43
Total uptime           : 4 years 26 weeks 0 days 11 hours 50 minutes
Total downtime         : 2 years 28 weeks 0 days 14 hours 48 minutes
Number of resets        : 116
Number of slot changes  : 0
Current reset reason    : PowerOn
Current reset timestamp : 12/22/2025 11:00:03
Current slot            : 1
Chassis type            : 0
Current uptime          : 0 years 7 weeks 1 days 21 hours 0 minutes
-----
```

```
-----
UPTIME CONTINUOUS INFORMATION
-----
```

Time Stamp	Reset	Uptime
MM/DD/YYYY HH:MM:SS	Reason	years weeks days hours minut
12/03/2025 11:43:20	Critical software exception	0 0 0 0 20
12/03/2025 11:57:40	Critical software exception	0 0 0 0 10
12/03/2025 12:18:12	Reload Command	0 0 0 0 15
12/04/2025 06:55:29	Reload Command	0 0 0 18 0
12/04/2025 07:38:17	Critical software exception	0 0 0 0 35
12/04/2025 07:54:40	Reload Command	0 0 0 0 10
12/04/2025 08:30:48	Critical software exception	0 0 0 0 30
12/04/2025 17:09:08	Image Install	0 0 0 8 0
12/05/2025 10:25:11	Reload Command	0 0 0 17 0
12/09/2025 15:36:41	Image Install	0 0 4 5 0
12/10/2025 16:22:01	Reload Command	0 0 1 0 0
12/10/2025 17:04:53	Critical software exception	0 0 0 0 35
12/10/2025 17:58:06	Critical software exception	0 0 0 0 45

5 subsequent unexpected reloads with 15 minutes or less uptime will disable autoboot

Crash files

- 2 type of files crash related files are stored in crashinfo.
Check also crashinfo-x: and stby-crashinfo:
- System reports , tar file, containing :
 - trace files
 - core file
 - various logs
- Crashinfo file, text file could contain:
 - partial crash info
 - Latest logs and commands executed
 - IOS Buffer pool info
 - Memory usage

File names contains
time stamp

```
R10_BN_19#dir stby-crashinfo:
18   98967374  Dec 14 2025 10:52:51 +00:00  R10_BN_19_1_RP_0-system-report_1_20251214-105237-UTC.tar.gz
17    479820  Dec 14 2025 10:51:32 +00:00  R10_BN_19_crashinfo_1_RP_00_00_20251214-105129-UTC
```


Packet Visibility

How to get insight into packet flows traversing the switch

- Netflow
Flow records send towards Netflow collectors
- SPAN/RSPAN/ERSPAN
Copies traffic towards packet capture devices
- NBAR/SDAVC
Overview of application flows
- Embedded Packet Capture
Copies traffic towards Wireshark process running on switch
- ELAM Capture
Captures hardware forwarding states of packets through switch

Switch Port Analyzer

- SPAN allows mirroring of traffic to a destination :
 - Local Destination (SPAN)
 - Remote through RSPAN vlan (RSPAN)
 - Remote using GRE encapsulation (ERSPAN)
- Mirrored packets do undergo queuing and scheduling on egress

```
Switch#show monitor session 1
Session 1
-----
Type                : ERSPAN Source Session
Status              : Admin Enabled
Source Ports        :
    Both            : Gi1/0/37
Destination IP Address : 10.48.13.181
MTU                 : 9000
Destination ERSPAN ID  : 1000
Origin IP Address    : 172.19.254.135
```

Wireshark · Packet 443746 · Ethernet0

```
> Frame 443746: 164 bytes on wire (1312 bits), 164 bytes captured (1312 bits) on interface \Device\NPF{...}
> Ethernet II, Src: Cisco_37:bf:c7 (2c:ab:eb:37:bf:c7), Dst: VMware_81:7c:62 (00:50:56:81:7c:62)
> Internet Protocol Version 4, Src: 172.19.254.135, Dst: 10.48.13.181
> Generic Routing Encapsulation (ERSPAN)
  Encapsulated Remote Switch Packet Analysis Type II
    0001 .... = Version: Type II (1)
    .... 0011 1111 1110 = Vlan: 1022
    000. .... = COS: 0
    ...1 1... = Encap: VLAN tag preserved in frame (3)
    .... 0... = Truncated: Not truncated (0)
    .... 11 1110 1000 = SpanID: 1000
    0000 0000 0000 .... = Reserved: 0
    .... 0000 0000 0000 0010 0101 = Index: 37
> Ethernet II, Src: Cisco_7c:4e:38 (54:8a:ba:7c:4e:38), Dst: Cisco_9f:fa:ef (00:00:0c:9f:fa:ef)
> Internet Protocol Version 4, Src: 172.19.1.9, Dst: 172.19.254.135
> Internet Control Message Protocol
```

Using Embedded Packet Capture

- EPC provides packet capture on Dataplane or control plane level
- Port Asic send copy of packets towards EPC process
- Limited in performance
- Multiple sessions can be configured but only one can be active

```
9300_54#monitor capture CL interface GigabitEthernet 1/0/11 in match any start
9300_54#monitor capture CL export location flash:cl.pcap
```

- Captures can be analysed off box, exported to a pcap file or displayed direct from the buffer
- Packet dissectors present on box to allow on box analysis and display
- Display filters present to allow further filtering

```
9300_54#show monitor capture file flash:cl.pcap display-filter arp
Starting the packet display ..... Press Ctrl + Shift + 6 to exit
 352  57.9 00:00:ca:fe:ca:fe ff:ff:ff:ff:ff:ff ARP 60 Who has 192.168.5.1? Tell 192.168.2.8
```

Verifying Hardware Forwarding (UADP platforms)

- Catalyst 9000 based switches allow for packets to be send through the system to verify hardware forwarding at that moment
- Show Platform Hardware switch <id> fed forward command allows building packet from CLI or through pcap to verify hardware forwarding
- Using “exact-route” provides a faster response but comes with less detailed information regarding forwarding path and decisions
- Run SPF with the *exact* details as packet ingresses. This should include a dot1q header if packet arrives on a trunk

Verifying Hardware Forwarding (UADP platforms)

```
9300_20#show platform hardware fed switch 1 forward exact-route interface te 1/0/1 0072.78f8.aed1
0072.78f8.1354 ipv4 10.48.91.151 10.48.13.151 tcp 10001 23 0
```

Ingress Packet Details:

```
ether hdr : dest mac: 0072.78f8.1354, src mac: 0072.78f8.aed1
ether hdr : ethertype: 0x0800 (IPv4)
ipv4  hdr : dest ip: 10.48.13.151, src ip: 10.48.91.151
ipv4  hdr : packet len: 40, ttl: 64, protocol: 6 (TCP)
tcp   hdr : dest port: 23, src port: 10001
```

Ingress Packet Hexdump (first 128 bytes):

```
007278F813550072 78F8AED608004500 0028000100004006 FD410A305B970A30
0D97271100170000 0000000000000500 2000EB2E00000000 0000000000000000
```

Egress Interface: **TenGigabitEthernet1/0/47**

Egress Packet Details:

```
ether hdr : dest mac: 0072.78f8.aed6, src mac: 0072.78f8.1355
ether hdr : ethertype: 0x0800 (IPv4)
ipv4  hdr : dest ip: 10.48.13.151, src ip: 10.48.91.151
ipv4  hdr : packet len: 40, ttl: 63, protocol: 6 (TCP)
tcp   hdr : dest port: 23, src port: 10001
```

Egress Packet Hexdump (first 128 bytes):

```
007278F8AED60072 78F8135508004500 0028000100003F06 FE410A305B970A30
0D97271100170000 0000000000000500 2000EB2E00000000 0000000000000000
```

Ingress Parameters

Egress packet

Verifying Hardware Forwarding (UADP platforms)

```
9300_20#show platform hardware fed switch 1 forward interface te 1/0/47 0072.78f8.aed1
0072.78f8.1354 ipv4 10.48.91.151 10.48.13.151 tcp 10001 23 0
Show forward is running in the background. After completion, syslog will be generated.
*Feb 10 14:52:48.864: %SHFWD-6-PACKET_TRACE_DONE: Switch 1 F0/0: fed: Packet Trace
Complete: Execute (show platform hardware fed switch <> forward last summary|detail)
```

Using packet captures with show platform forward reduces complexity

```
9300_20#show platform hardware fed switch 1 forward interface te 1/0/47 pcap
flash:inpacket.pcap number 2 data
```

Results:

- show platform hardware fed switch <id> forward last summary
- show platform hardware fed switch <id> forward last detail
- shfwd.log files in flash:

Control Plane Policing HW stats

- Control plane policing drops frames on port ASIC's to protect CPU
- Do not modify defaults without clear understanding of the impact

9300_1#show plat hardware fed switch active qos queue stats internal cpu policer

CPU Queue Statistics

QId	PlcIdx	Queue Name	Enabled	(default) Rate	(set) Rate	Queue Drop (Bytes)	Queue Drop (Frames)
4	2	Routing Control	Yes	5400	5400	0	0
5	14	Forus Address resolution	Yes	4000	65000	2212622868	345698644

CPU Queue Policer Statistics

Policer Index	Policer Accept Bytes	Policer Accept Frames	Policer Drop Bytes	Policer Drop Frames
12	0	0	0	0
13	9562218	122704	453808601	7079824

CPP Classes to queue map

PlcIdx	CPP Class	Queues
0	system-cpp-police-data	ICMP GEN/ BROADCAST/ ICMP Redirect/
14	system-cpp-police-forus	Forus Address resolution/ Forus traffic/

Control Plane policing drops

Rate changed from Default rate

Determining Cause of Inband traffic

9300_54#sh platform software fed switch active punt rates interfaces

Packets per second averaged over 10 seconds, 1 min and 5 mins

Active interfaces
sending to cpu

		Recv	Recv	Recv	Drop	Drop	Drop
		10s	1min	5min	10s	1min	5min
Interface Name	IF_ID						
GigabitEthernet1/0/11	0x00000013	5999	1707	2074	0	0	0
Vlan192	0x00000036	5999	1707	2074	0	0	0

9300_54#sh platform software fed switch active punt cpuq rates

Packets per second averaged over 10 seconds, 1 min and 5 mins

Per Queue Statistics

		Rx	Rx	Rx	Drop	Drop	Drop
		10s	1min	5min	10s	1min	5min
Q no	Queue Name						
0	CPU_Q_DOT1X_AUTH	0	0	0	0	0	0
1	CPU_Q_L2_CONTROL	0	0	0	0	0	0
5	CPU_Q_FORUS_ADDR_RESOLUTION	6000	1874	2414	0	0	0

Queue
information

Drops typically seen on Port
Asic not in FED punt path

Determine packets hitting a specific CPU queue

- FED allows limited packet captures done in punt and inject path
- Allows filter to be setup to collect specific data.
- Use filter “fed.queue==<queue>” for collecting specific queue
other filters , eth.addr, ip.addr , etc
- Detail and brief packet display.
- Detail contains internal forwarding information

```
9300_54#debug plat soft fed switch active punt packet-capture set-filter "fed.queue == 5"
9300_54#debug platform software fed switch active punt packet-capture [start|stop]
9300_54#show platform software fed switch active punt packet-capture brief
Punt packet capturing: disabled. Buffer wrapping: disabled
Total captured so far: 4096 packets. Capture capacity : 4096 packets
Capture filter : "fed.queue == 5"
----- Punt Packet Number: 3, Timestamp: 2023/05/31 13:14:45.888 -----
interface : physical: GigabitEthernet1/0/11[if-id: 0x00000013], pal: Vlan192 [if-id: 0x00000036]
metadata  : cause: 7 [ARP request or response], sub-cause: 1, q-no: 5, linktype: MCP_LINK_TYPE_IP
ether hdr  : dest mac: ffff.ffff.ffff, src mac: 0000.cafe.cafe
ether hdr  : ethertype: 0x0806 (ARP)
```

Metadata indicates
why it was punted

Packet Drops

Checking interface status

show interface status errdisable
or show log for cause

Vlan column shows routed(L3) ,
trunk or base vlanid

Half Duplex 1Gb/s, possible
duplex mismatch?

```
9500_1#show interfaces status | ex notc
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Twe1/0/7		connected	routed	a-full	a-1000	10/100/1000BaseTX SFP
Twe1/0/12		err-disabled	1	full	10G	SFP-10GBase-CU1M
Twe1/0/15		connected	routed	full	10G	SFP-10GBase-CU3M
Twe1/0/20		connected	4094	full	10G	SFP-10GBase-CU1M
Twe1/0/21		connected	routed	a-half	a-1000	10/100/1000BaseTX SFP
Twe1/0/22		disabled	1	auto	auto	10/100/1000BaseTX SFP
Hu1/0/26		connected	trunk	full	40G	QSFP 40G CU3M
Hu1/0/27		connected	4094	full	100G	QSFP 100G CU2M

Show interface status command gives a quick overview of interface status

Verifying link utilization

```
Switch#show interfaces | inc line|rate
```

```
Vlan1 is up, line protocol is up , Autostate Enabled
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

```
GigabitEthernet0/0 is administratively down, line protocol is down
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

```
GigabitEthernet1/0/11 is up, line protocol is up (connected)
```

```
30 seconds input rate 575000 bits/sec, 958 packets/sec
```

```
30 seconds output rate 126975000 bits/sec, 10473 packets/sec
```

```
GigabitEthernet1/0/12 is down, line protocol is down (notconnect)
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

```
GigabitEthernet1/0/13 is down, line protocol is down (notconnect)
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

Input/output rates show average over 5 minutes (default). Traffic might be bursty in nature minimum value is 30 seconds

Verifying link utilization

Input/output rates show average over 5 minutes (default). Traffic might be bursty in nature minimum value is 30 seconds

```
Switch#show interfaces | inc line|rate
Vlan1 is up, line protocol is up , Autostate Enabled
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
GigabitEthernet0/0 is administratively down, line protocol is down
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
GigabitEthernet1/0/11 is up, line protocol is up (connected)
  30 seconds input rate 575000 bits/sec, 958 packets/sec
  30 seconds output rate 126975000 bits/sec, 10473 packets/sec
GigabitEthernet1/0/12 is down, line protocol is down (notconnect)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
GigabitEthernet1/0/13 is down, line protocol is down (notconnect)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
```

```
Switch#show controllers utilization
Port          Receive Utilization  Transmit Utilization
Gi1/0/1       0                    0
Gi1/0/11      0                    92
.
Gi1/0/24      0                    0
Te1/1/1       0                    0
Te1/1/4       9                    0
Total Ports   : 33

Total Ports Receive Bandwidth Percentage Utilization : 0
Total Ports Transmit Bandwidth Percentage Utilization : 0
Average Switch Percentage Utilization : 0
```

Bandwidth in %
Current load

Ethernet Statistics

```
Switch#show controllers ethernet-controller gi 5/0/48
```

```
Transmit
```

```
1562496684 Total bytes
 5032561 Unicast frames
700808558 Unicast bytes
1269484 Multicast frames
861688062 Multicast bytes
  1 Broadcast frames
  0 Cos 0 Pause frames
1236978 Minimum size frames
1892419 65 to 127 byte frames
1941967 128 to 255 byte frames
 685594 256 to 511 byte frames
 20261 512 to 1023 byte frames
524827 1024 to 1518 byte frames
  0 8192 to 16383 byte frames
  0 16384 to 32767 byte frame
  0 > 32768 byte frames
  0 Late collision frames
  0 Excess Defer frames
  0 Good (1 coll) frames
  0 Good (>1 coll) frames
  0 Deferred frames
```

```
LAST UPDATE 361 msec AGO
```

```
Receive
```

```
2968958225 Total bytes
 6004241 Unicast frames
1807110661 Unicast bytes
 2789759 Multicast frames
1161847500 Multicast bytes
  1 Broadcast frames
  0 Cos 0 Pause frames
 871517 Minimum size frames
2181611 65 to 127 byte frames
2712229 128 to 255 byte frames
1260418 256 to 511 byte frames
 900135 512 to 1023 byte frames
868091 1024 to 1518 byte frames
  0 8192 to 16383 byte frames
  0 16384 to 32767 byte frame
  0 > 32768 byte frames
  0 SymbolErr frames
  0 Collision fragments
  0 ValidUnderSize frames
  0 InvalidOverSize frames
  0 ValidOverSize frames
```

Ethernet controller statistics give
more detailed port statistics
then show interface

Frame size
distribution

Error statistics

Tail Drops

```
9300_54#show interfaces gigabitEthernet 1/0/11 | inc output drops
  Input queue: 0/2000/0/0 (size/max/drops/flushes); Total output drops: 1277
9300_54#show controllers ethernet-controller gig 1/0/11 | inc Excess Def
      1277 Excess Defer frames                0 Collision fragments
SNMP:
SNMPv2-SMI::enterprises.9.2.2.1.1.27.8 = INTEGER: 1277
```

- Tail drops occur when more traffic queues up and reaches one of the thresholds on the asic or no buffers are available
- Queueing occurs when going from higher to lower bandwidth links.
40Gb/s in , 1Gb/s out could result in 39 packets needing to be buffered for while 1 packet send
- Buffer allocation per class can be changed inside service-policy associated to the interfaces
- Global multiplier to increase buffers up to 1200%
qos softmax-queue-multiplier <percentage>
Only adjust when needed, drops are not always a problem
- Not all traffic is equal, file transfers don't notice buffering, voice/video/transactional does

QoS Hardware configuration

- Qos Configuration performed using policy maps that are applied per interface
- Show policy-map displays statistics per class
- Use "auto qos .." cli or Catalyst Centre to ease deployment of QoS policies

```
9500_1#sh run | sec DNA-dscp#APIC_QOS_Q_OUT
policy-map DNA-dscp#APIC_QOS_Q_OUT
  class DNA-EZQOS_2P6Q3T_9K#VOICE-PQ1
    priority level 1
    police rate percent 10
    queue-buffers ratio 15
  class DNA-EZQOS_2P6Q3T_9K#VIDEO-PQ2
    priority level 2
    police rate percent 23
    queue-buffers ratio 10
  class DNA-EZQOS_2P6Q3T_9K#CONTROL-PLANE
    bandwidth remaining percent 10
    queue-buffers ratio 10
!
interface TwentyFiveGigE1/0/2
  service-policy output DNA-dscp#APIC_QOS_Q_OUT
```

```
9500_1#show policy-map interface tw 1/0/2
TwentyFiveGigE1/0/2
Service-policy output: DNA-dscp#APIC_QOS_Q_OUT
  queue stats for all priority classes:
    Queueing
    priority level 1
    (total drops) 0
    (bytes output) 0
  queue stats for all priority classes:
    Queueing
    priority level 2
    (total drops) 0
    (bytes output) 0
Class-map: DNA-EZQOS_2P6Q3T_9K#VOICE-PQ1 (match-any)
  3994 packets
Match:  dscp ef (46)
Priority: Strict,
Priority Level: 1
police:
  rate 10 %
  rate 2500000000 bps, burst 78125000 bytes
conformed 0 bytes; actions:
  transmit
exceeded 0 bytes; actions:
  drop
```

QoS Hardware configuration

- Limited buffers are reserved per port, most buffers come from the global pool
- Shaping occurs through queuing , policers are handled through hardware policers
- Hardware supports multiple Queues with multiple thresholds per queue
- When exceeding threshold traffic corresponding traffic will be dropped

```
9300_45#sh plat hard fed switch active qos queue config interface gi 1/0/11
Asic:0 Core:1 DATA Port:11 GPN:1 LinkSpeed:0x1
```

DTS		Hardmax		Softmax		PortSMin		GlblSMin		PortStEnd	
0	1	5	200	12	3200	5	500	0	0	6	9600
1	1	4	0	13	4800	5	750	2	300	6	9600
Priority		Shaped/shared		weight		shaping_step		sharpedWeight			
0	0	Shared		50		0		0			
1	0	Shared		75		0		0			
Port		Port		Port		Port		Port			
Priority		Shaped/shared		weight		shaping_step					
2		Shaped		254		255					
Weight0		Max_Th0	Min_Th0	Weigth1	Max_Th1	Min_Th1	Weight2	Max_Th2	Min_Th2		
0	0	2709		0	0	3028	0	0	3400	0	
1	0	3825		0	0	4275	0	0	4800	0	

Hardmax : Reserved
Softmax : Global pool

Queue mode:
shaped or shared
Queue limit for shaping
Step/weight * speed

Drop thresholds for
queue/threshold
In Buffers (256 byte)

QoS hardware statistics

```
9300_45#sh platform hardware fed switch active qos queue stats interface gigabitEthernet 1/0/11
```

AQM Global counters

```
GlobalHardLimit: 7976 | GlobalHardBufCount: 0
GlobalSoftLimit: 11872 | GlobalSoftBufCount: 0
```

Highest number of buffers in use by interface since last issuing command

High Watermark Soft Buffers: 429 <--- clear on read

Asic:0 Core:1 DATA Port:10 Hardware Enqueue Counters

Q	Buffers (Count)	Enqueue-TH0 (Bytes)	Enqueue-TH1 (Bytes)	Enqueue-TH2 (Bytes)	Qpolicer (Bytes)
0	200	0	385820	46085690	0
1	0	0	0	0	0

Current buffer count per queue

Asic:0 Core:1 DATA Port:0 Hardware Drop Counters

Q	Drop-TH0 (Bytes)	Drop-TH1 (Bytes)	Drop-TH2 (Bytes)	SBufDrop (Bytes)	QebDrop (Bytes)	QpolicerDrop (Bytes)
0	0	0	412312	0	0	0
1	0	0	0	0	0	0

- At Asic level there are 8 Queues/3 Thresholds
- Enqueue/Drop Counters available per queue/per threshold
- Buffers (count) show currently assigned buffers to Queue (256 bytes)
- To enable high water mark counter monitoring.
set platform hardware fed switch active qos port-monitor interface <if>

Doppler ASIC packet forwarding drop counters

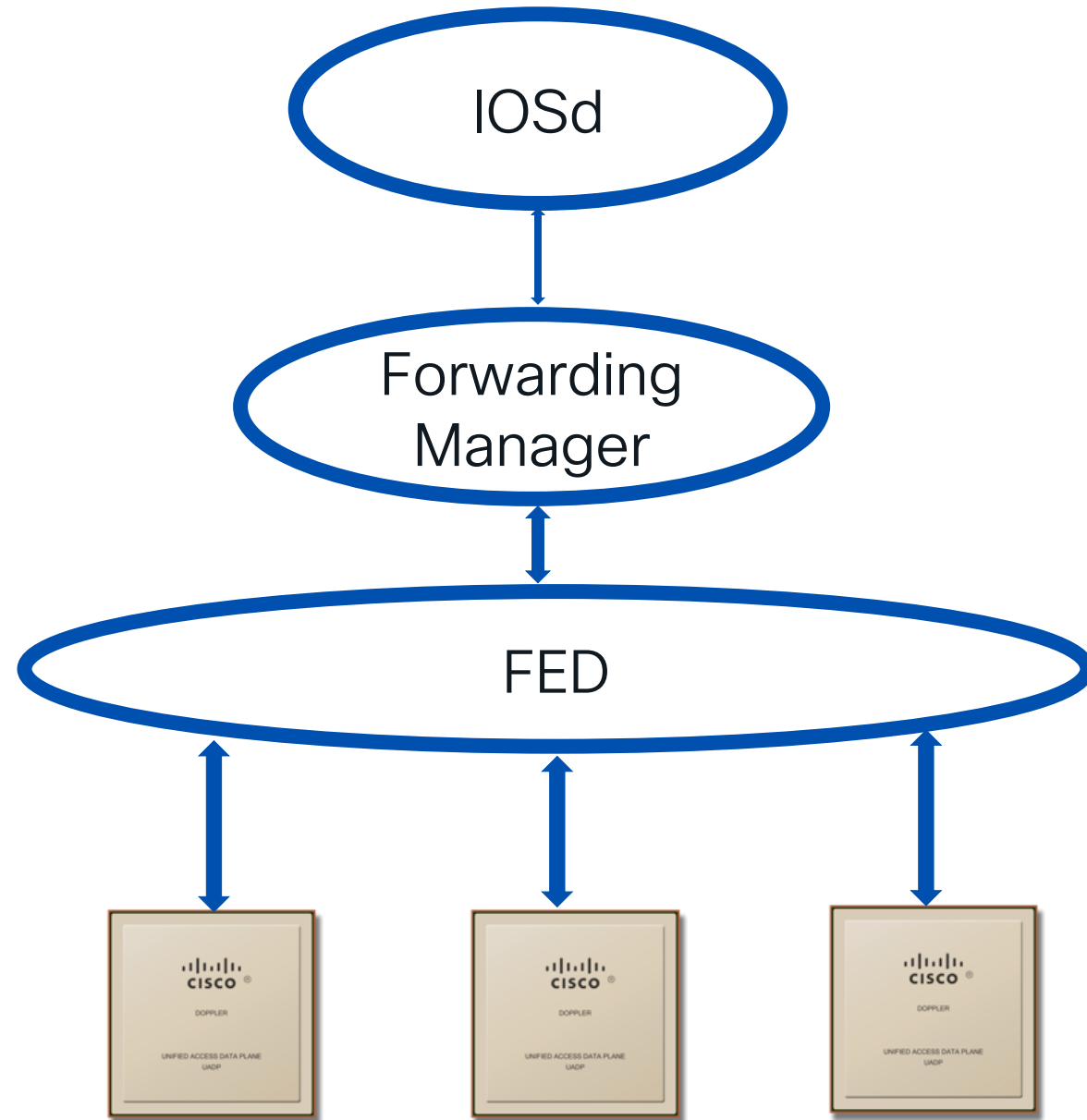
```
9300_45#sh platform hardware fed switch active fwd-asic drops exceptions
***EXCEPTION STATS ASIC INSTANCE 0 (asic/core 0/0)***
=====
Asic/core | NAME | prev | current | delta
=====
0 0 NO_EXCEPTION 35364016 35364108 92
0 0 IPV4_CHECKSUM_ERROR 0 0 0
0 0 ROUTED_AND_IP_OPTIONS_EXCEPTION 2 2 0
0 0 CTS_FILTERED_EXCEPTION 0 0 0
0 0 AUTH_DRIVEN_DROP 0 0 0
0 0 PKT_DROP_COUNT 0 3732 3732
0 0 ALLOW_DOT1Q_EXCEPTION_COUNT 0 0 0
0 0 ALLOW_PRIORITY_TAGGED_EXCEPTION_COUNT 0 0 0
0 0 IGR_EXCEPTION_L5_ERROR 0 363 363
0 0 IP_UNICAST_TTL_REACHED_ZERO 0 0 0
0 0 MISC_FATAL_ERROR 0 0 0
```

- Every packet passing through Port Asic gets parsed by the port-asic's on both receive and transmit side
- Exception drops are counted per Asic, not per port.

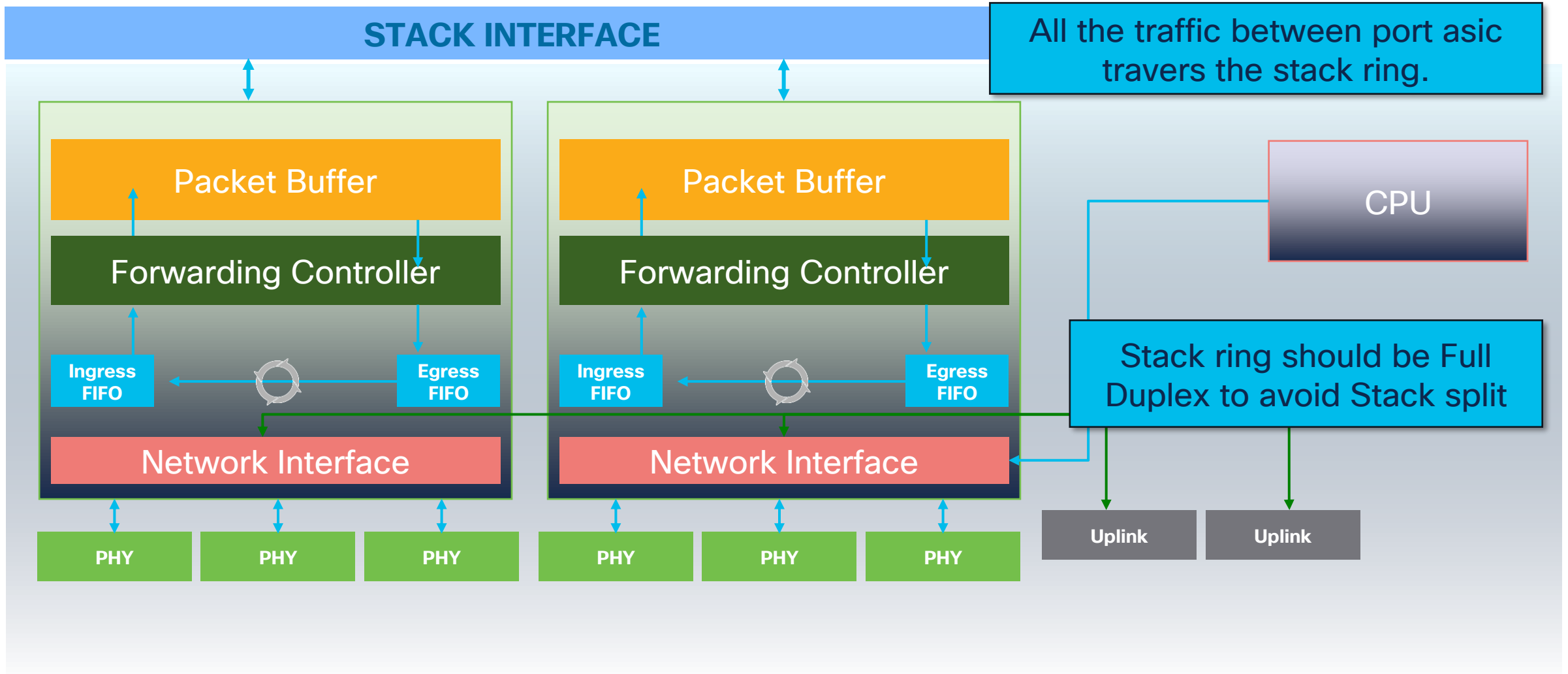
Forwarding Verification

Troubleshooting Forwarding

- Many layers are involved in forwarding
- Troubleshoot from top (IOSd) down toward the hardware layer
- 9500X and 9600 (Supervisor 2) use Silicon One
All other models based on UADP architecture
- Packet processing happens on port asics.
- CPU only involved in programming the asic for forwarding.
- Understand the packet flow through the system



Catalyst 9200/9300 Stackable Switches

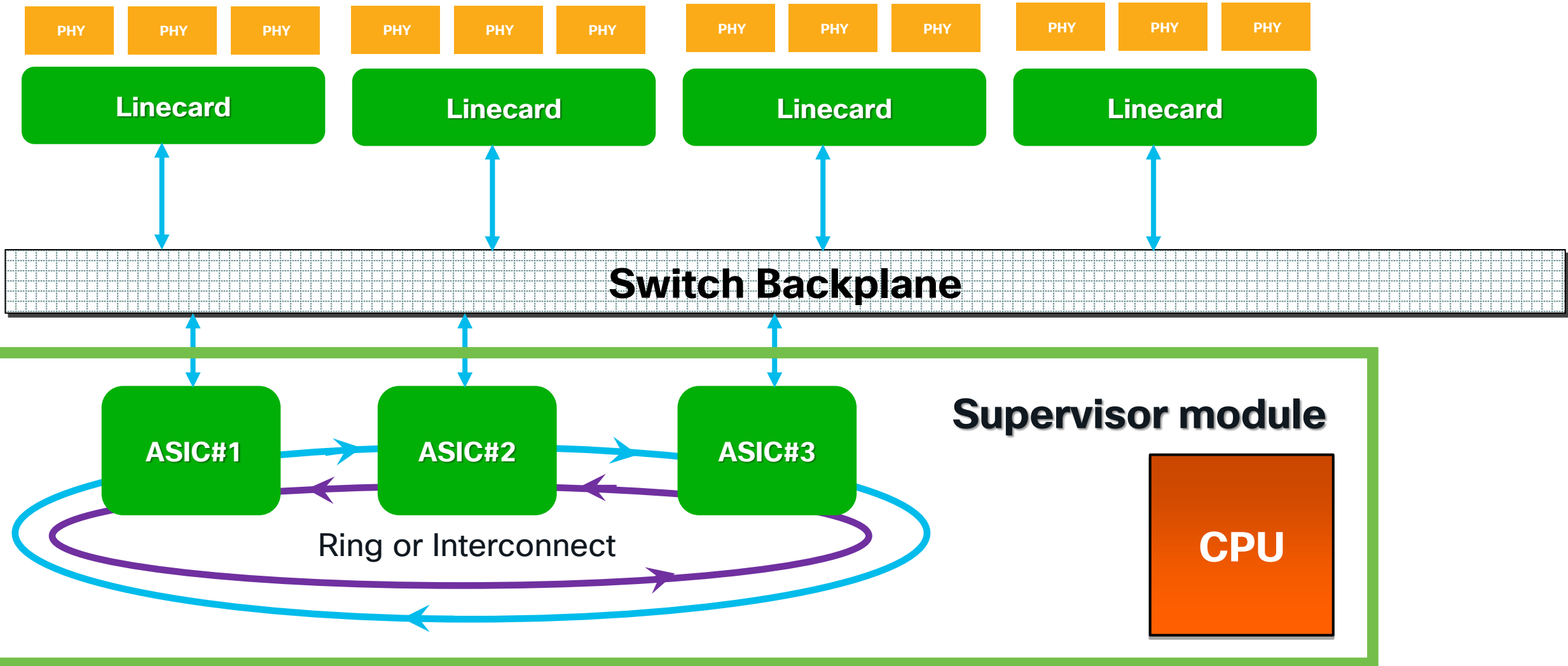


Switch Stacks

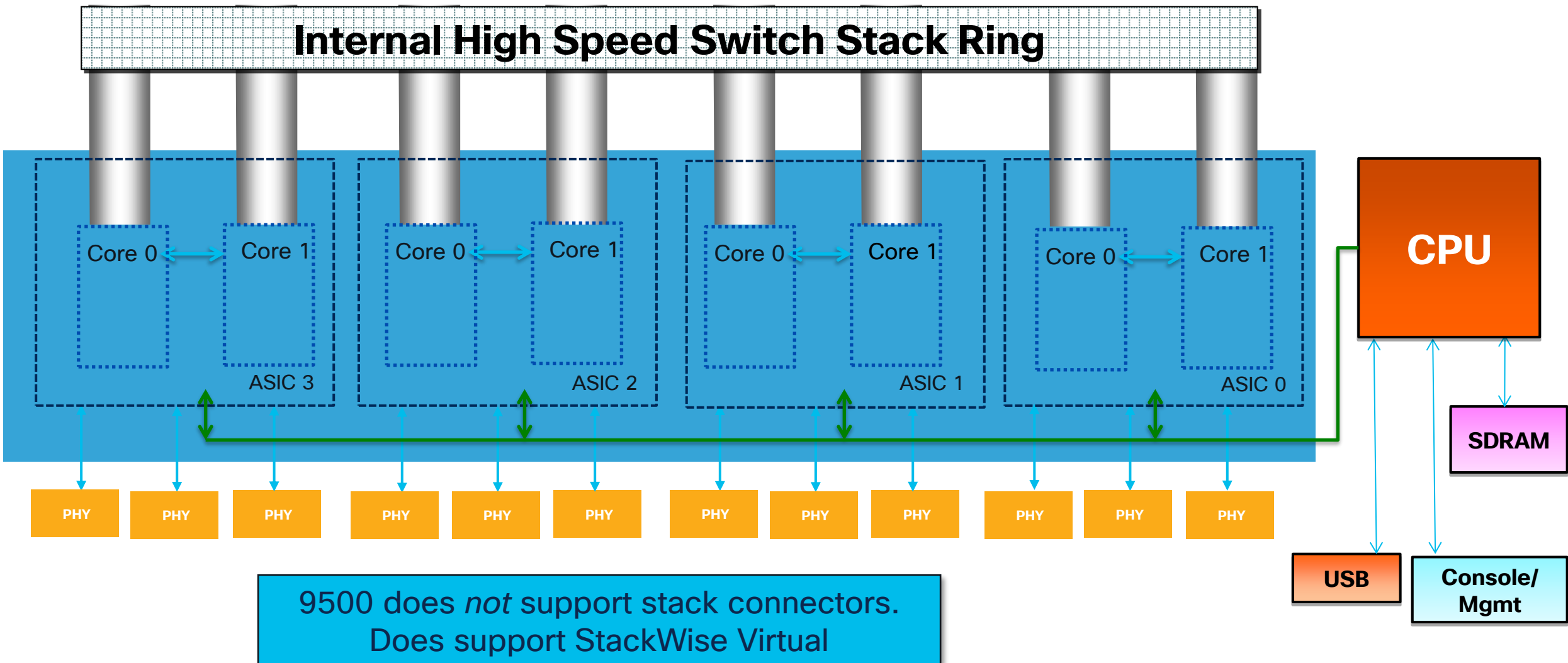
9300_10#sh switch stack-ports summary									
Sw#/Port#	Port Status	Neighbor/Port	Cable Length	Link OK	Link Active	Sync OK	#LinkOK	In Loopback	
1/1	OK	4/2	100cm	Yes	Yes	Yes	67557	No	
1/2	OK	2/1	50cm	Yes	Yes	Yes	1	No	
2/1	OK	1/2	50cm	Yes	Yes	Yes	2	No	
2/2	OK	3/1	50cm	Yes	Yes	Yes	1	No	
3/1	OK	2/2	50cm	Yes	Yes	Yes	1	No	
3/2	OK	4/1	50cm	Yes	Yes	Yes	2	No	
4/1	OK	3/2	50cm	Yes	Yes	Yes	1	No	
4/2	OK	1/1	100cm	Yes	Yes	Yes	2	No	

- Ensure all stack ports are connected and Link Ok to ensure stack is running with redundancy and full bandwidth
- High Number of #linkOK or inability to read cable length might indicate bad cable/stack ports
- Show tech-support stack to gather useful commands

Catalyst 9400/9600 Chassis based



Catalyst 9500 Fixed Switches



FED Interface Internal Mappings

Interface to ASIC mapping important to understand data flows

```
9300_1#show platform software fed switch active ifm mappings
```

Interface	IF_ID	Inst	Asic	Core	Port	SubPort	Mac	Cntx	LPN	GPN	Type	Active
GigabitEthernet1/0/1	0x8	1	0	1	0	0	26	6	1	1	NIF	Y
GigabitEthernet1/0/2	0x4c	1	0	1	1	0	6	7	2	2	NIF	Y
GigabitEthernet1/0/3	0x4d	1	0	1	2	0	28	8	3	3	NIF	Y

Internally used interface addressing:

- LPN : Local Port Number
- GPN : Global Port Number
- IF_ID : Interface Identification, used for many fed CLI
- Type : Type of interface, NIF = Network Interface
- Inst : Instance : ASIC + Core
- Port : Asic Ports
- Active : Is Interface Active

Troubleshoot between ports on same ASIC when possible

FED IFM Mappings logical interface

```
9600_SVL#show platform software fed switch active ifm interfaces vlan
Interface                IF_ID                State
-----
Vlan1                    0x0005ffff          READY
Vlan123                  0x00420012          READY
Vlan900                  0x00420010          READY
9600_SVL#show platform software fed switch active ifm interfaces svi
Interface                IF_ID                State
-----
Vlan123                  0x0000013b          READY
9600_SVL#show platform software fed switch active ifm interfaces tunnel
Interface                IF_ID                State
-----
Tunnell                  0x00000143          READY
```

- An unique IF_ID is assigned to every logical and physical.
- Every type of interface has its unique IF_ID.
Ex: Layer 2 Vlan has a different IF_ID as the Layer 3 SVI

FED IF_ID more detail

```
9600_SVL#show platform software fed switch active ifm if-id 0x138
```

```
Interface IF_ID      : 0x00000000000000138
Interface Name       : FortyGigabitEthernet2/1/0/1
Interface Block Pointer : 0x7f33b89fd918
Interface Block State  : READY
Interface State       : Enabled
Interface Status      : ADD, UPD
Interface Type        : ETHER
Port Type             : ROUTE PORT
Port Location        : LOCAL
Slot                 : 15
Unit                 : 0
Slot Unit            : 1
SNMP IF Index        : 148
GPN                  : 577
Port Handle          : 0xb7000104
IPv4 MTU             : 9100
IPv6 MTU             : 0
IPv4 VRF ID          : 0x0
IPv6 VRF ID          : 0x0
Protocol flags       : 0x0007 [ ipv4 ipv6 pim_ipv4 ]
```

Specifying the IF-ID gives
verbose information

Information available varies
depending on Interface Type

One VRF ID per VRF, VRF ID
0 is Global Routing Table

Layer 2 Forwarding. Verifying HW STP state

- Show spanning tree gives IOSd view of Spanning Tree
- Hardware forwarding states can be checked *per switch* on FED layer
- Outputs will show what interface are in forwarding state and if traffic is tagged or untagged
- Flood list indicates what Ports will receive flooded traffic on this switch

```
9300_54#sh platform hardware fed switch active vlan 192 egress
VLAN STP State in hardware
vlan id is:: 192
Interfaces in forwarding state: : Gi1/0/11(Untagged) , Te1/1/1(Untagged) , Te1/1/4(Tagged)
9300_54#sh platform hardware fed switch active vlan 192 ingress
VLAN STP State in hardware
vlan id is:: 192
Interfaces in forwarding state: : Gi1/0/11(Untagged) , Te1/1/1(Untagged) , Te1/1/4(Tagged)
flood list: : Gi1/0/11, Te1/1/1, Te1/1/4
```

Layer 2 Forwarding, IOSd mac address tables

```
9300_54#show mac address vlan 192
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
192    0000.cafe.cafe     DYNAMIC   Gi1/0/11
192    0013.c3c1.0d89     DYNAMIC   Gi1/0/11
192    0050.5693.5e70     DYNAMIC   Te1/1/4
192    a0f8.4910.2dd3     STATIC    V1192
192    d0ec.35c9.d353     DYNAMIC   Te2/1/1
Total Mac Addresses for this criterion: 5
```

- Show mac address-table command shows overview of all mac addresses dynamic, static and system mac addresses.
- IOSd shows global view.
Each FED maintains its own Mac Address Table

FED MATM Mac Address Table

```
9300_1#sh platform software fed switch 1 matm macTable vlan 100
```

VLAN	MAC	Type	Seq#	EC_Bi	Flags	machandle	siHandle	diHandle	*a_time	*e_time	ports
192	0000.cafe.cafe	0x1	71718	0	0	0x7f2348	0x7f2198	0x7f2478	300	112	GigabitEthernet1/0/11
192	d0ec.35c9.d353	0x101	71719	0	0	0x7f2328	0x7f2148	0x7f2588	300	14	TenGigabitEthernet2/1/1
192	0013.c3c1.0d89	0x1	71720	0	0	0x7f23d8	0x7f2378	0x7f4278	300	11	GigabitEthernet1/0/11
192	a0f8.4910.2dd3	0x8002	0	0	64	0x7f2308	0x7f21c8	0x5234	0	0	Vlan192
192	0050.5693.5e70	0x1	12772	0	0	0x7f23d8	0x7f2358	0x7f23e8	300	13	TenGigabitEthernet1/1/4

```
*a_time=aging_time(secs) *e_time=total_elapsed_time(secs)
```

```
Type:
```

MAT_DYNAMIC_ADDR	0x1	MAT_STATIC_ADDR	0x2	MAT_CPU_ADDR	0x4	MAT_DISCARD_ADDR	0x8
MAT_ALL_VLANS	0x10	MAT_NO_FORWARD	0x20	MAT_IPMULT_ADDR	0x40	MAT_RESYNC	0x80
MAT_DO_NOT_AGE	0x100	MAT_SECURE_ADDR	0x200	MAT_NO_PORT	0x400	MAT_DROP_ADDR	0x800
MAT_DUP_ADDR	0x1000	MAT_NULL_DESTINATION	0x2000	MAT_DOT1X_ADDR	0x4000	MAT_ROUTER_ADDR	0x8000
MAT_WIRELESS_ADDR	0x10000	MAT_SECURE_CFG_ADDR	0x20000	MAT_OPQ_DATA_PRESENT	0x40000	MAT_WIRED_TUNNEL_ADDR	0x80000
MAT_DLR_ADDR	0x100000	MAT_MRP_ADDR	0x200000	MAT_MSRRP_ADDR	0x400000	MAT_LISP_LOCAL_ADDR	0x800000
MAT_LISP_REMOTE_ADDR	0x1000000	MAT_VPLS_ADDR	0x2000000				

- Type Field indicates the type of mac address using bitmap.
- Sequence number of an entry changing would indicate relearning
- Owning FED responsible for aging entries. MAT_DO_NOT_AGE bit set in type for non owning FED instances

Layer 3 Forwarding. Routing protocols

```
9300_20#sh ip route 10.48.13.0
Routing entry for 10.48.13.0/24
  Known via "isis", distance 115, metric 30, type level-2
  Redistributing via isis
  Last update from 172.18.255.9 on TenGigabitEthernet1/0/47, 00:40:35 ago
  Routing Descriptor Blocks:
    * 172.18.255.9, from 172.31.254.18, 00:40:35 ago, via TenGigabitEthernet1/0/47
      Route metric is 30, traffic share count is 1

9300_20#show ip arp 172.18.255.9
Protocol  Address            Age (min)  Hardware Addr   Type   Interface
Internet  172.18.255.9        41         0072.78f8.aed6  ARPA   TenGigabitEthernet1/0/47
```

- Verify routing table and next hop information with known correct path
- Confirm the route shown is covered by the correct subnet and learned by the expected routing protocol.

Cisco Express Forwarding

- CEF shows the Forwarding Information Base showing how traffic will be forwarded through the system.
- Sources show where the route is learned from.
RIB(Routing Information Base)
- Use Detail or Internal keywords to get even more detail

```
9300_20#show ip cef 10.48.13.0 internal
10.48.13.0/24, epoch 4, RIB[I], refcnt 6, per-destination sharing
  sources: RIB
    feature space:
      IPRM: 0x00028000
      Broker: linked, distributed at 4th priority
    ifnums:
      TenGigabitEthernet1/0/47(144): 172.18.255.9
    path list 70C9760F41F0, 327 locks, per-destination, flags 0x4D [shble, hvsh, rif, hwcn]
      path 70C978450268, share 1/1, type attached nexthop, for IPv4
        nexthop 172.18.255.9 TenGigabitEthernet1/0/47, IP adj out of TenGigabitEthernet1/0/47, addr
172.18.255.9 70C9761536E8
    output chain:
      IP adj out of TenGigabitEthernet1/0/47, addr 172.18.255.9 70C9761536E8
```

Cisco Express Forwarding

```
9300_20#show adjacency 172.18.255.9 detail
Protocol Interface Address
IP TenGigabitEthernet1/0/47 172.18.255.9(174)
0 packets, 0 bytes
sourced in sev-epoch 9
Encap length 14
007278F8AED6007278F813550800
L2 destination address byte offset 0
L2 destination address byte length 6
Link-type after encap: ip
ARP
9300_20#sh ip arp TenGigabitEthernet 1/0/47
Protocol Address Age (min) Hardware Addr Type Interface
Internet 172.18.255.10 - 0072.78f8.1355 ARPA TenGigabitEthernet1/0/47
Internet 172.18.255.9 221 0072.78f8.aed6 ARPA TenGigabitEthernet1/0/47
```

- CEF adjacency displays next hop information.
- Layer 2 rewrite information displayed to verify Destination and Source Mac Address of rewritten packet

Forwarding Manager (FP/RP)

- Forwarding Manager should show the prefix on both Standby/Active RP and on all FP (stack members or supervisor modules)
- aom id shown with FP can be used to display prefix in object manager

```
9300_20#sh platform software ip switch active F0 cef prefix 10.48.13.0/24 detail
```

```
Forwarding Table
```

```
10.48.13.0/24 -> OBJ_ADJACENCY (0x4a), urpf: 81
```

```
Prefix Flags: unknown
```

```
aom id: 11562925, HW handle: (nil) (created)
```

```
9300_20#sh platform software object-manager switch active f0 object 11562925
```

```
Object identifier: 11562925
```

```
Description: PREFIX 10.48.13.0/24 (Table id 0)
```

```
Obj type id: 71
```

```
Obj type: route-pfx
```

```
Status: Done, Epoch: 0, Client data: 0x452c6ae8
```

Forwarding Manager (FP/RP)

- Forwarding Manager should show the prefix on both Standby/Active RP and on all FP (stack members or supervisor modules)
- AOM id shown with FP can be used to display prefix in Object Manager

```
9300_20#show platform software adjacency switch active f0 index 0x4a
Adjacency id: 0x4a (74)
  Interface: TenGigabitEthernet1/0/47, IF index: 144, Link Type: MCP_LINK_IP
  Encap: 0:72:78:f8:ae:d6:0:72:78:f8:13:55:8:0
  Encap Length: 14, Encap Type: MCP_ET_ARPA, MTU: 9100
  Flags: no-l3-inject
  Nexthop addr: 172.18.255.9
  IP FRR MCP_ADJ_IPFRR_NONE 0
  aom id: 8386846, HW handle: (nil) (created)
```

Forwarding Manager (FP/RP)

- Looking at downlinks, parents or children of an object helps traverse the data for related objects

```
9300_20#show platform software object-manager switch active f0 object 8386846
Object identifier: 8386846
  Description: adj 0x4a, Flags None
  Obj type id: 40
  Obj type: adj
  Status: Done, Epoch: 0, Client data: 0x4546afd8
9300_20#sh plat soft object-manager switch active f0 object 8386846 downlinks
Object identifier: 987
  Description: PREFIX 0.0.0.0/0 (Table id 0)
  Status: Done
Object identifier: 11562925
  Description: PREFIX 10.48.13.0/24 (Table id 0)
  Status: Done
..
```

Error Objects

- Error object that do not get cleared could suggest forwarding issues
- Lookup objects to verify relationship to potential issues

```
9600X_50#show platform software object-manager f0 statistics
```

```
Forwarding Manager Asynchronous Object Manager Statistics
```

```
Object update: Pending-issue: 2, Pending-acknowledgement: 0
```

```
Batch begin: Pending-issue: 0, Pending-acknowledgement: 0
```

```
Batch end: Pending-issue: 0, Pending-acknowledgement: 0
```

```
Command: Pending-acknowledgement: 0
```

```
Total-objects: 1897
```

```
Stale-objects: 0
```

```
Resolve-objects: 0
```

```
Childless-delete-objects: 0
```

```
Backplane-objects: 0
```

```
Error-objects: 9
```

```
Number of bundles: 0
```

```
Paused-types: 0
```

Error Objects

- Use keyword error-object, pending-* to see object id's
- Check objects in error to confirm their origin
- Error objects are an error but not always result in traffic loss

```
9600X_50#sh platform software object-manager f0 error-object
Object identifier: 650830
  Description: LB 0x10000417
  Status: Modify/delete-failed
9600X_50#sh platform software object-manager f0 object 650830
Object identifier: 650830
  Description: LB 0x10000417
  Obj type id: 60
  Obj type: loadbalance
  Status: Modify/delete-failed, Epoch: 0, Client data: 0xad0635e8
```


FED Routing tables

```
9300_20#show plat software fed switch active ip route 10.48.13.0/24 detail
vrf      dest                htm                flags      SGT      DGID Last-modified                SecsSinceHit
---      ----                ---                -----      ---      ----
0        10.48.13.0/24          0x78bcb94983a8 0x0         0        0      2025/02/04 20:36:50.190          3
FIB: prefix_hdl:0x2600007a, mpls_ecr_prefix_hdl:0, sgtOverWrite: 0
===== OCE chain =====
ADJ:objid:74 {link_type:IP ifnum:0x90, adj:0x62000050, si: 0x78bcb9199178 IPv4: 172.18.255.9 }
[Output omitted]
Station Index (SI) [0xc3]
RI = 0x1f
DI = 0x53a8
Replication Bitmap: LD
Destination index   = 0x53a8
pmap                = 0x00000000 0x400000000000
pmap_intf : [TenGigabitEthernet1/0/47]
..
```

- FED layer output dependant on platform used (S1 vs Doppler)
- Replication Bitmap (per Asic/Core) traffic is Local, Remote or other Core.
- Bits in pmap are set if traffic would go out an interface on that asic

FED Adjacency table

```
9300_20#sh platform software fed switch 1 ip adj 172.18.255.9 detail
IPV4 Adj entries
dest          if_name          dst_mac          si_hdl  ri_hdl  adj_id  Last-modified
172.18.255.9  TenGiga1/0/47  0072.78f8.aed6   0x78178 0x78158 0x4a    2024/12/31 16:31:09
Asic          SI-Index          DI-Index
----          -
0             195              0x53a8
Destination index = 0x53a8
pmap          = 0x00000000 0x40000000000000
pmap_intf : [TenGigabitEthernet1/0/47]
Rewrite info:
  ASIC#:0 RI:31 Rewrite_type:AL_RRM_REWRITE_L3_UNICAST_IPV4_SHARED(1)
Mapped_rii:L3_UNICAST_IPV4(9)
          MAC Addr: MAC Addr: 00:72:78:f8:ae:d6,
          L3IF LE Index 38
```

- Rewrites occur on destination Asic
- Check Adj on Egress switch (when in stack)
- Traffic can also be send towards RCP (Recirculation Path)

FED Tables, Silicon One

```
9600X_50#show plat software fed active ip route 10.48.13.0/24
IPV4ROUTE_ID:id:0x5dd182def298 nobj:(IPNEXTHOP_ID,0x1000001d) 10.48.13.0/24 tblid:0 DA:0
  NPD: device:0 lsparec:0 api_type:route(3)
  SDK: is_host:0 l3_dest:0x7b8bedf76590 l3_dest:la_l3_fec_impl_base(oid=2851) vrf(gid/oid):0/0x267
  ADJ:objid:0x1000001d (IPv4: 172.31.250.218) nh_type:NHADJ_NORMAL iif_id:0x4bc ether_type:0x8
#child:105
  srcmac:a453.0e46.22c5 dstmac:4ce1.75eb.0e82
  NPD: fec_oid:2851 was_nor_nh:1 cr_def:0 stale:0 l3port_valid:1
  NPD: device:0 nh_gid/oid:10/0xb1a old_gid/oid:0/0x0 parent_oid:2711
  SDK: cla_nh_type:0
9600X_50#show plat software fed active ip adj 172.31.250.218
Number of IPv4 Adjacency entries: 13
Adjacency ID      Adjacency IPAddress      Dest MacAddr      Nexthop Type      Interface
0x1000001d        172.31.250.218          4ce1.75eb.0e82    NORMAL            HundredGigE1/0/16
```

- Silicon One outputs differ from Doppler based platform on FED layer but can provide similar outputs
- Note : Not all FED commands work on both Doppler and S1.

TCAM route utilization

```
9300_20#sh platform software fed switch active ip route summary
Total number of v4 fib entries = 11348
Total number succeeded in hardware = 8186

Mask-Len 0 :- Total-count 1 hw-installed count 1
Mask-Len 4 :- Total-count 1 hw-installed count 1
Mask-Len 8 :- Total-count 2 hw-installed count 2
Mask-Len 24 :- Total-count 10002 hw-installed count 8171
Mask-Len 30 :- Total-count 1331 hw-installed count 0
Mask-Len 32 :- Total-count 11 hw-installed count 11
```

- Hardware installed count should match Total-Count
- Routes not installed into Hardware are mostly due to resource issues
- TCAM's use regions to install routes , might not achieve 100% efficiency

Verifying TCAM Utilization

```
9300_20#show platform hardware fed switch active fwd-asic resource tcam utilization
```

```
Codes: EM - Exact_Match, I - Input, O - Output, IO - Input & Output, NA - Not Applicable
```

```
CAM Utilization for ASIC [0]
```

Table	Subtype	Dir	Max	Used	%Used	V4	V6	MPLS	Other
Mac Address Table	EM	I	32768	97	0.30%	0	0	0	97
Mac Address Table	TCAM	I	1024	21	2.05%	0	0	0	21
L3 Multicast	EM	I	8192	0	0.00%	0	0	0	0
L3 Multicast	TCAM	I	512	9	1.76%	3	6	0	0
L2 Multicast	EM	I	8192	0	0.00%	0	0	0	0
L2 Multicast	TCAM	I	512	11	2.15%	3	8	0	0
IP Route Table	EM	I	24576	12	0.05%	11	0	1	0
IP Route Table	TCAM	I	8192	8190	99.98%	8177	10	2	1
QOS ACL	TCAM	IO	5120	181	3.54%	52	86	0	43
Security ACL	TCAM	IO	5120	129	2.52%	26	58	0	45
Netflow ACL	TCAM	I	256	42	16.41%	20	20	0	2

- TCAM Full situation can lead to performance issues
- Choose suitable SDM template to optimize TCAM allocation
- TCAM utilization can vary per ASIC
- Check syslog for error messages

Before needing to troubleshoot

- Get familiar with the CLI's before you need them
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